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SHORT COMMUNICATION

Ghost Net Entanglement of a Vulnerable Seahorse *Hippocampus kelloggi* (Teleostei; Syngnathidae) from Dhamara Estuary, Bay of Bengal, India

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Abstract Present study describes entanglement of a sea horse *Hippocampus kelloggi* in discarded nylon fishing nets or "Ghost nets". This was observed during the routine sampling and monitoring program on 25th August 2021 in Dhamara estuary, Bay of Bengal, Odisha. This specimen is different from other similar species having well developed coronate with five short spines, backward pointing rounded cheek spine; rounded eye spine; thick body rings; thick snout and narrow body. Ghost net entanglement imposes pressure to *Hippocampus kelloggi* which is under vulnerable (VU) category in the IUCN Red List of Threatened Species. Therefore, it is decisive to initiate appropriate management steps to regulate the abandoned fishing nylon net in the coastal and marine ecosystem to protect the marine biodiversity.

Keywords Ghost net entanglement · *Hippocampus kelloggi* · Dhamara estuary · Conservation and management · Bay of Bengal

Abandoned fishing nets also known as "Ghost nets", can be described as discarded fishing nets; damaged fishing nets or accidentally abandoned fishing gear that are lost in the sea or ocean during fishing activity [1–4]. Such ghost nets are otherwise called as "abandoned, lost or otherwise discarded fishing gear" (ALDFG), by which both targeted and non-targeted marine fauna become the victim. Ghost net entanglement of marine organisms leads to severe physical damage that further leads to mortality. Therefore, such contemporaneous issues are recently getting momentum from the scientific fraternity [1, 4]. Recent studies describe that more than 5400 marine megafaunal communities that have entangled in or associated with ghost fishing gear such as elasmobranchs, reptiles, mammals, sea turtles, etc. [2, 3]. It is well noticed that ghost nets affect marine life excessively, as a result of which quantum loss of marine life becomes a huge challenge to measure and document [2]. As per estimation, more than 6.4 million tonnes of ghost nets are released into the seas and oceans every year [4]. However, estimation reveals around 6% of most of the fishing nets, 9% of all fish traps and above 29% of all lines getting vanished in the oceans per annum [1]. This value might be too low while comparing with the real scenario. Most of the gill nets became a menace for the coastal and marine fishery resources, since they are related to unintended by catch of fish and shellfish fauna.

Sea horse fishing is basically done by the small scale fishers for the aquarium trade, traditional medicines and curios. The demand of dried sea horse for the preparation of Traditional Chinese Medicine (TCM) is at boom. Most of the sea horse caught by the deep sea trawl nets and gill nets accidentally [5]. Therefore, both targeted catch and bycatch are being associated with sea horse fishery, run parallel to the market demand. The depletion of wild populations and impoverishment from the native habitat becomes a common phenomenon that represents catastrophic effects of anthropogenic stress reasonable impede in a cascade manner.

There are 41 extant species of *Hippocampus* confirmed after the molecular tool based taxonomic revision [6] published in 2016. However, according to the recent IUCN database there are 46 extant species of sea horse residing in the

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global marine waters; out of which 12 species are grouped under vulnerable (VU), 17 species are data deficient (DD), 10 species are least concerned (LC), 4 species are not evaluated (NE), 2 species are endangered (EN) and 1 species is near threatened (NT) (source: https://www.iucn-seahorse.org/seahorses-seadragons). Mostly sea horse species are confined to the tropical belt of Indo-Pacific region. All sygnathidian species are protected under the Indian Wildlife Protection Act, 1972 under Schedule—I (Part IIA). All Seahorses were listed on the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) enlisted in Appendix II in 2002, with implementation as of May 2004.

Along the Indian waters 9 species of seahorse are predominated out of 12 species preponderant of the Indo-Pacific region (Table 1). These nine species are broadly distributed in both the east and west coast of India including the Gulf of Mannar, Andaman and Nicobar waters. The most preferable habitats for sea horse are mangroves, coral beds, macro algal beds, sea grass beds, coastal wetlands, estuarine ecosystems and creek ecosystems. Recently, migration of sea horse has been recorded from the south east coast towards northeast coast of India [7]. This is likely due to the massive fishing pressure, loss of feeding and breeding grounds coupled with augmented anthropogenic stress [7].

In India much research has been done on beach litter pollution [8, 9]. However, ghost net entanglement of fauna has gotten the least scientific attention [10]. Contemporary published research literature reveals that sea horse entanglement in ghost fishing net from mangrove belt has not been recorded so far from the Odisha coast. Under this backdrop, the present study describes the occurrence of a seahorse, found entangled in a ghost net from Dhamara estuary, India.

During the regular sampling program of Coastal Management Cell (CMC) of State Pollution Control Board, Odisha at Dhamara estuary and adjoining nearshore waters (Fig. 1), the survey team observed and recorded dead seahorse species entangled in the ghost net (Fig. 2A) on 25th August 2021. In situ photographs were taken by the GPS enabled digital mobile hand set with photographic camera lens. The specimen was collected carefully without making any physical damage and preserved in a 10% formalin solution. Then the sample was transported to the laboratory for the taxonomic identification. Subsequently, morphological characters were measured to identify the specimen up to species level. The taxonomic identification is made by following the standard literatures [6, 11]. The study area map (Fig. 1)was prepared by ArcGIS 10.2 software.

The observed specimen possesses the distinctive morphological characters such as 40 tail rings, 11 trunk rings; 2 rings supporting the dorsal fin, trunk rings and 1 tail ring; 18 dorsal fin rays and 17 pectoral fin rays. Coronet is high, having five short spines; spine structure is low and rounded; cheek spine is long slightly backwards-pointing, deep head with narrow body; conspicuous body rings; round but small eye; snout thick; female, length 18.5 cm (Fig. 2 B).

After considering all these features, the specimen was identified as *Hippocampus kelloggi* Jordan and Snyder 1902, having the common name "Kellogg's seahorse" or "great sea horse" with the following classification.

Class: Teleostei.

Order: Syngnathiformes.

Table 1 Seahorse distribution along Indian coast

Sl. no.	Seahorse species	Common name	Location	References
1	*H. comes* Cantor, 1849	Tiger-tail Seahorse, tiger-tailed seahorse	Andaman Islands	Lourie et al. [6]
2	*H. histrix* Kaup, 1856	Thorny Seahorse	Both the east and west coast of India	Lourie et al. [6]
3	*H. kelloggi* Jordan and Snyder, 1901	Great Seahorse, Kellogg's seahorse, offshore seahorse	Both the east and west coast of India	Lourie et al. [6]
4	*H. kuda* Bleeker, 1852	Spotted Seahorse, common seahorse, estuarine seahorse, oceanic seahorse, yellow seahorse	Both the east and west coast of India	Lourie et al. [6]
5	*H. mohnikei* Bleeker, 1853	Japanese Seahorse	East coast of India	Lourie et al. [6]
6	*H. spinosissimus* Weber, 1913	Hedgehog Seahorse	East coast of India	Lourie et al. [6]
7	*H. trimaculatus* Leach, 1814	Three-spot Seahorse, flat-faced seahorse, longnose seahorse, low-crowned seahorse, smooth seahorse,	Both the east and west coast of India	Lourie et al. [6]
8	*H. camelopardalis* Bianconi, 1854	Giraffe sea horse	Gulf of Kachchhh Waters, North west coast of India	Subburaman et al. [15]
9	*H. montebelloensis* Kuiter, 2001	Monte Bello seahorse	Gulf of Mannar, South east coast of India	Yogeshkumar et al. [16]

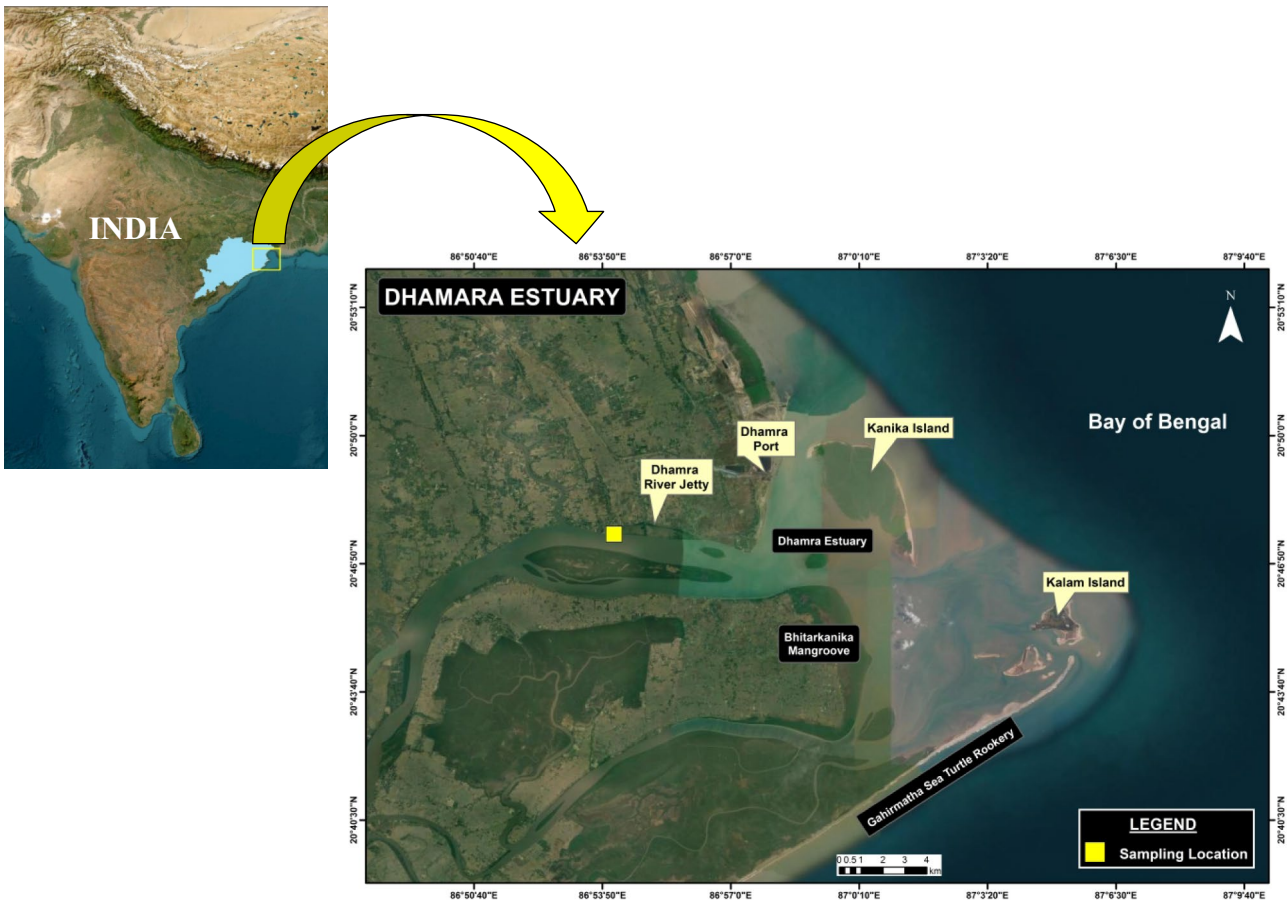

Fig. 1 Map showing the location of *Hippocampus kelloggi* entangled in abandoned fishing net at Dhamara estuary, India

Fig. 2 **A** Abandoned nylon fishing net (Ghost Net) entanglement of *Hippocampus kelloggi* in the Dhamara estuary, India and **B** lateral view of *Hippocampus kelloggi*

Family: Syngnathidae Bonaparte, 1831.

Genus: *Hippocampus* Rafinesque, 1810.

Species: *Hippocampus kelloggi* Jordan and Snyder, 1902 (Fig. 2B).

Material examined—Registration Number—(State Pollution Control Board/Coastal Management Cell) SPCB/CMC-Seahorse-25/08/2021, Dhamara estuary, Odisha, Bay of Bengal, 20° 47′ 34″ N and 86° 54′ 07″ E (Fig. 2A and B).

Distribution- Indo-Pacific: India, China, Indonesia, Japan, Malaysia, Pakistan, Philippines, Thailand, United Republic of Tanzania, Vietnam [6, 11].

Body colouration- pale yellow and uniform in colour.

This species is different from other contemporary species such as *H. spinosissimus* which bears a deep body; a few tail rings, few pectoral fin rays. Further, *H. algiricus*, has fewer tail rings and broad (almost double) eye and cheek spines, but its distribution is restricted to West Africa only. *H. kelloggi* also resembles *H. ingens,* but different in having fewer pectoral fin rays and more dorsal fin rays [11].

Occurrence of seahorse in the north east coast of India is not common [6, 7, 11]. Earlier published documents [7] suggest that seahorse *Hippocampus fuscus* later synonymized as *Hippocampus kuda* is confined to Gopalpur to Rushikulya estuary stretched about 40 km, followed by Chilika Lake. Seahorse migration through the East India Coastal Current (EICC) has been reported, which covers a distance of more than 1100 km ranging from the south east coast to the north east coast of India [7]. Further observations are made, describing the importance of ocean currents and fronts in population structures of the *Hippocampus mohnikei* complex in Indian context [12]. In general *H. kelloggi* is a non-migratory, bathydemersal species however its migration may be possible through passive mode of transport. For instance, migration of *Hippocampus kelloggi* through the East Australian Current has been documented, that ranges from the north Australia coast to the south Australia coast through the East Australian Current (EAC) covering 900 km [13]. Similarly, in the Indian context migration of *H. kelloggi* from south east coast towards north east coast of India (south Odisha coast) covering 1300 km has been reported [14]. However, this migration of seahorses may be supported by a unique method called rafting, where small seahorse species attach themselves to floating objects like macroalgae or plastic debris and are dispersed by the ocean currents, such as the east Indian coastal current and north-east and south-west monsoon currents [14]. Such long range migration of seahorse *H. kelloggi* through coastal currents might be a forcing factor to entanglement within the ghost fishing nets or ALDFG.

Recently, new distributional records of seahorse along Indian coasts have been reported such as *Hippocampus camelopardalis* [15] from the Gulf of Kachchh; north west coast of India and *Hippocampus montebelloensis* [16] from the southeast coast of India. This reflects that the coastal waters of India are highly conducive for seahorse survival in spite of augmented plastic pollution in the coastal and nearshore marine ecosystems. Certainly, marine plastic pollution and ghost nylon fishing nets put additional stress upon their longevity.

The occurrence of *Hippocampus kelloggi* in Dhamara mangrove fringe, northwestern Bay of Bengal has been recorded for the first time. Attachment of seahorse *H. kelloggi* in abandoned, lost or otherwise discarded fishing gear (ALDFG) seems challenging for the survival of such a faunal group. Since, there is no information available on the quantification of seahorse with ghost fishing nylon nets in the Bay of Bengal region, a real assessment of marine organism's entanglement is found difficult to address. Proper marine policy framework is needed to control and quantify the ghost net as well ghost net entangled organisms for better management to protect the marine biodiversity from the edge of degradation.

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Declarations

Conflict of interest The authors have no conflict of interest to declare.

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